

DOM-Based XSS Vulnerability

The **Document Object Model (DOM)** is a programming interface for web documents, representing the page structure as a tree of objects where JavaScript can manipulate to change content, structure, or style dynamically. A **JavaScript DOM vulnerability** occurs when client-side scripts insecurely process attacker-controllable data, leading to unintended behavior or security breaches.

The key subtype is **DOM-based XSS** (also called Type-0 XSS), where malicious input from a controllable source (like a URL fragment) is passed to a "sink" function that executes or renders it, allowing arbitrary JavaScript to run in the victim's browser without server involvement. Unlike other XSS types, the payload never reaches the server—it's all client-side.

Other related DOM vulnerabilities include:

- **DOM Clobbering**: Injecting HTML to override JavaScript variables or functions, altering app behavior.
- **DOM-based Open Redirects** or **Cookie Manipulation**: Using sinks to redirect users or tamper with cookies.

These fall under client-side injection in the OWASP Top 10.

History

Past:

XSS vulnerabilities date back to the late 1990s with the rise of dynamic web pages. Early mentions appeared in 1997 discussions about malicious HTML in discussion forums, but the term "Cross-Site Scripting" was coined around 2000 by Microsoft engineers. DOM-based XSS was formally identified and named by security researcher Amit Klein in 2005, as client-side JavaScript grew more complex with AJAX and rich interfaces. Before this, most XSS was server-side (reflected or stored).

Early exploits targeted simple sinks like `eval()` or `innerHTML`. Notable cases:

- 2011: jQuery plugins had DOM XSS bugs allowing script injection.
- 2013: Gawker Media (Lifehacker, Gizmodo) and Alexa (Amazon subsidiary) had DOM XSS flaws enabling account hijacks.
- High-profile sites like Google, Yahoo, and Amazon reported DOM XSS issues, affecting up to 50% of websites by the 2010s.

Detection relied on manual code review or basic tools like `grep` for patterns like `innerHTML = location.hash`.

Present:

Today, with single-page applications (SPAs) dominant, DOM vulnerabilities are widespread. Claranet's 2024 report found over 2,570 XSS instances in 500 pentests, many DOM-based. Tools have evolved for automated detection and ethical exploitation:

- **Detection Tools:**
 - Burp Suite (with DOM Invader extension): Scans for sources/sinks and simulates attacks.
 - OWASP ZAP: Free DAST tool for injecting payloads into client-side code.
 - Acunetix: High detection rate for DOM XSS via DeepScan technology.
 - eslint-plugin-no-unsanitized: Static analysis for JavaScript code to flag unsafe sinks like innerHTML.
 - Browser DevTools: Manual interception of DOM events.
- **Exploitation Tools:**
 - Manual payloads via URL manipulation (e.g., `#<script>alert(1)</script>`).
 - DOM Invader or Untrusted Types: Automate source/sink tracing.

Ethical hackers use these in pentests to demonstrate impact without harm.

Future:

By 2026 and beyond, trends point to increased prevalence due to AI-generated code, which introduces XSS in 86% of cases. Supply chain attacks via vulnerable libraries (e.g., recent jQuery CVE-2020-11023 added to CISA catalog in 2025) will rise. However, frameworks like React 19+ emphasize secure defaults, server-first rendering, and AI-assisted vulnerability detection. Expect more integration of static analysis in IDEs and AI tools to flag DOM issues early. With WebAssembly and microfrontends, new DOM manipulation risks may emerge, but CSP and secure coding will mitigate them.

Uses in Ethical Hacking

In ethical hacking, DOM vulnerabilities are tested to uncover client-side weaknesses:

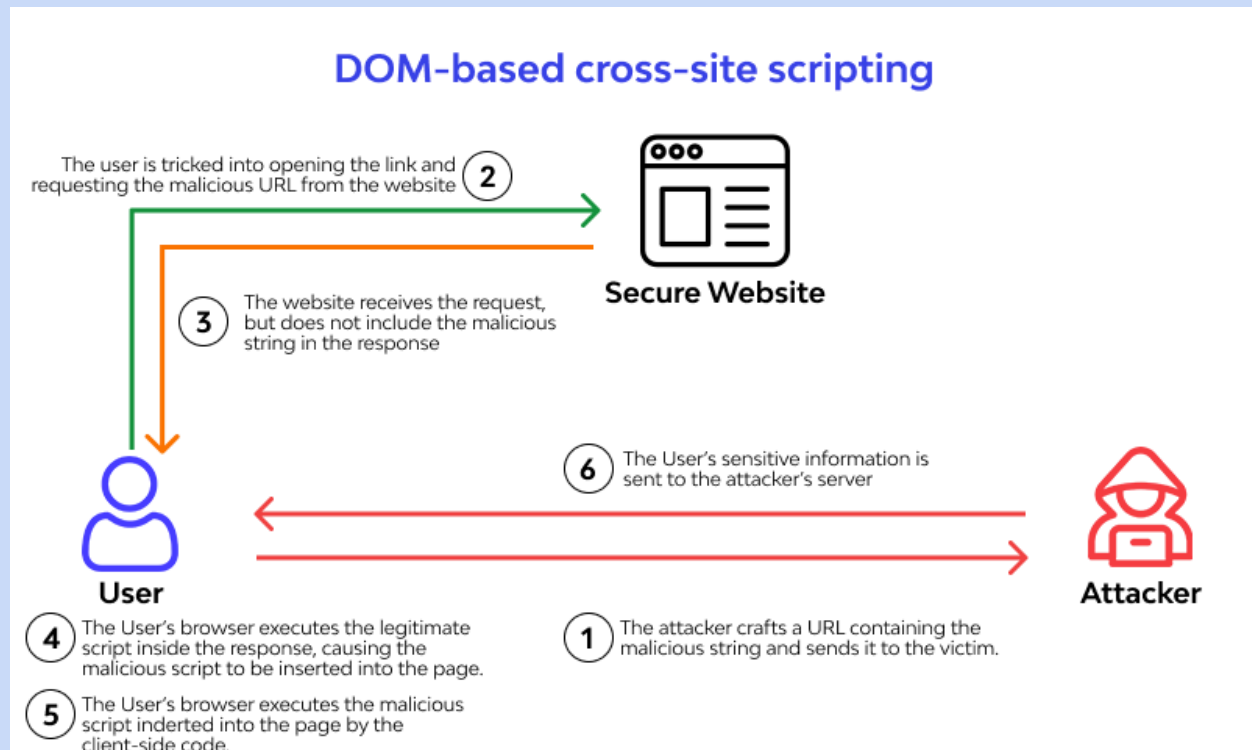
- **Pentesting:** Demonstrate how attackers could steal sessions or deface pages.
- **Bug Bounties:** Platforms like HackerOne reward DOM XSS findings.
- **Red Teaming:** Simulate phishing with crafted URLs to bypass server filters.
- **Defensive:** Train developers on secure JS practices, like using safe sinks (e.g., `textContent` instead of `innerHTML`).

Not for malicious use—always get permission.

Working Procedure

DOM-based XSS follows a "source-to-sink" flow:

1. **Source:** Attacker controls input (e.g., `location.hash`, `document.cookie`, `localStorage`).
2. **Processing:** JS reads the source without sanitization (e.g., `var userInput = location.hash.slice(1);`).
3. **Sink:** Input is passed to a dangerous function (e.g., `document.getElementById('output').innerHTML = userInput;`), executing the payload.
4. **Execution:** Browser runs the malicious JS, like stealing cookies via `document.cookie`.



This diagram illustrates the attack flow: Attacker crafts a URL, victim loads it, browser executes the payload client-side.

Differences with Other Vulnerabilities

Vulnerability Type	DOM-based XSS	Reflected XSS	Stored XSS
Location	Client-side only (browser DOM manipulation)	Server reflects input in response	Server stores input, serves to users
Payload Delivery	Via URL fragments, cookies, etc.; no server interaction	Via query params; server echoes back	Via forms/comments; persisted in DB
Persistence	Non-persistent (per page load)	Non-persistent	Persistent (affects all viewers)
Detection Difficulty	High (requires client-side scanning)	Medium (check responses)	Low (scan stored data)
Examples	<code>innerHTML</code> from <code>location.hash</code>	Server echoes unsanitized GET param	Malicious comment displayed to all
Mitigation Focus	Client-side sanitization (e.g., DOMPurify)	Server output encoding	Server input validation + encoding

Example: Basic DOM XSS (Beginner)

localhost/dvwa/vulnerabilities/xss_d/?default=Spanish



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Brute Force

Command Injection

CSRF

File Inclusion

File Upload

Insecure CAPTCHA

SQL Injection

SQL Injection (Blind)

Weak Session IDs

XSS (DOM)

XSS (Reflected)

XSS (Stored)

CSP Bypass

JavaScript Attacks

Authorisation Bypass

Vulnerability: DOM Based Cross Site Scripting (XSS)

Please choose a language:

Spanish Select

More Information

- <https://owasp.org/www-community/attacks/xss/>
- https://owasp.org/www-community/attacks/DOM_Based_XSS
- <https://www.acunetix.com/blog/articles/dom-xss-explained/>

localhost/dvwa/vulnerabilities/xss_d/?default=raja



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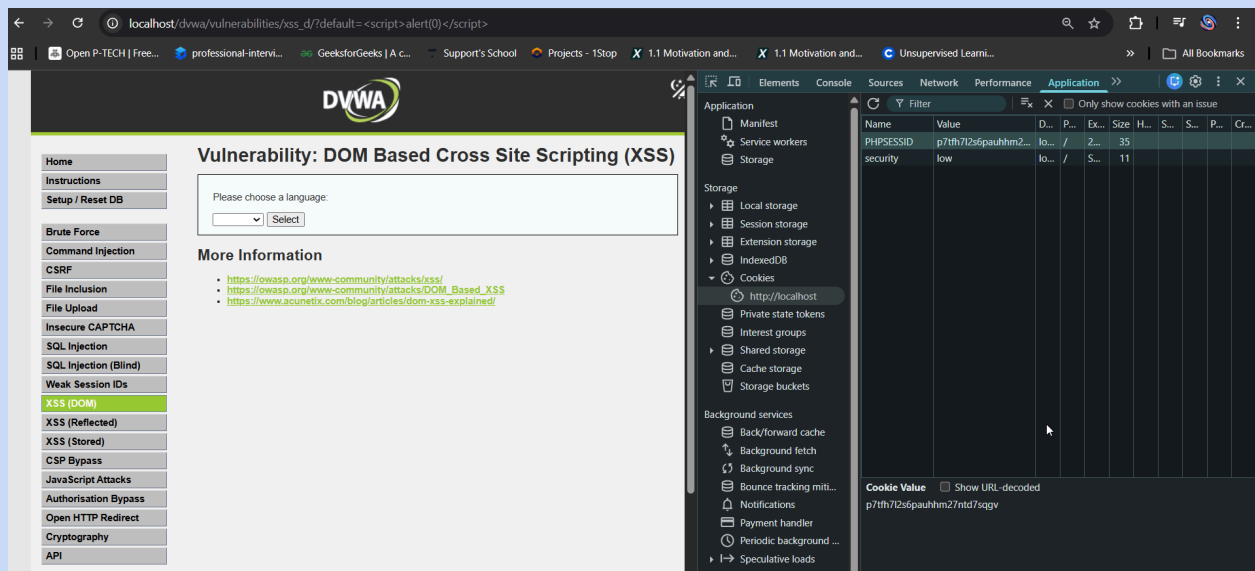
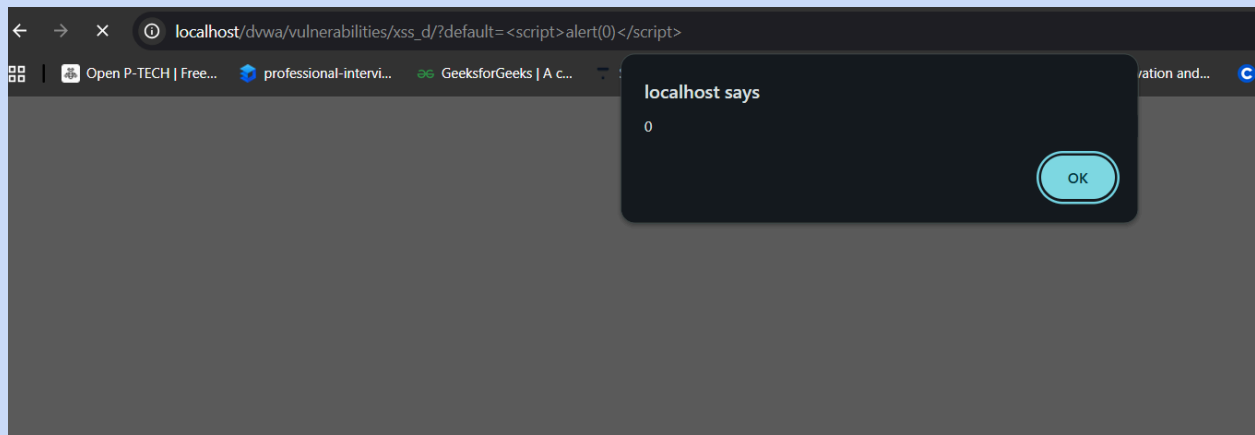
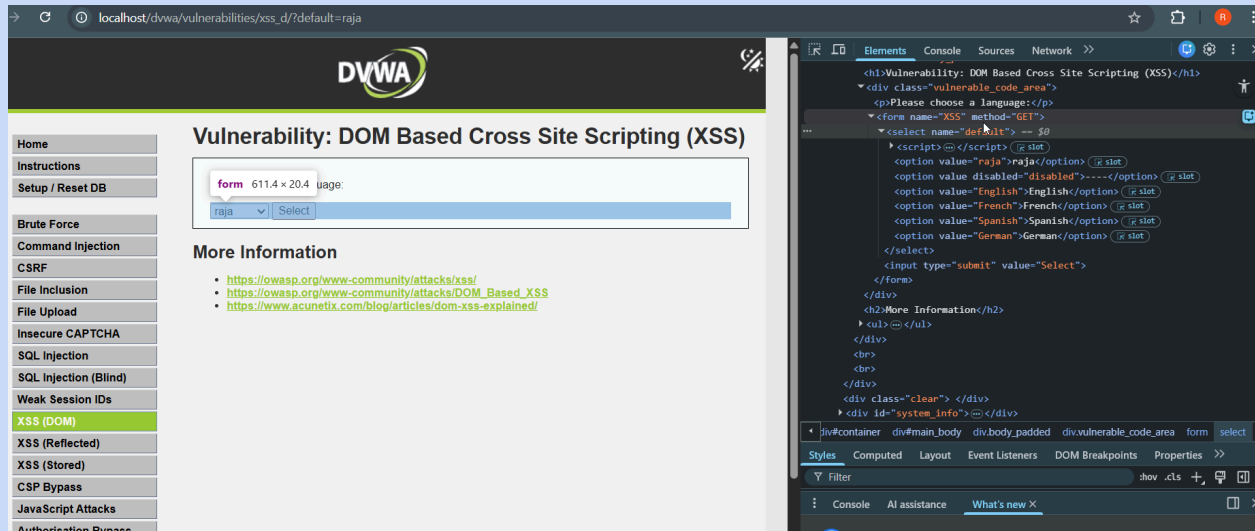
Vulnerability: DOM Based Cross Site Scripting (XSS)

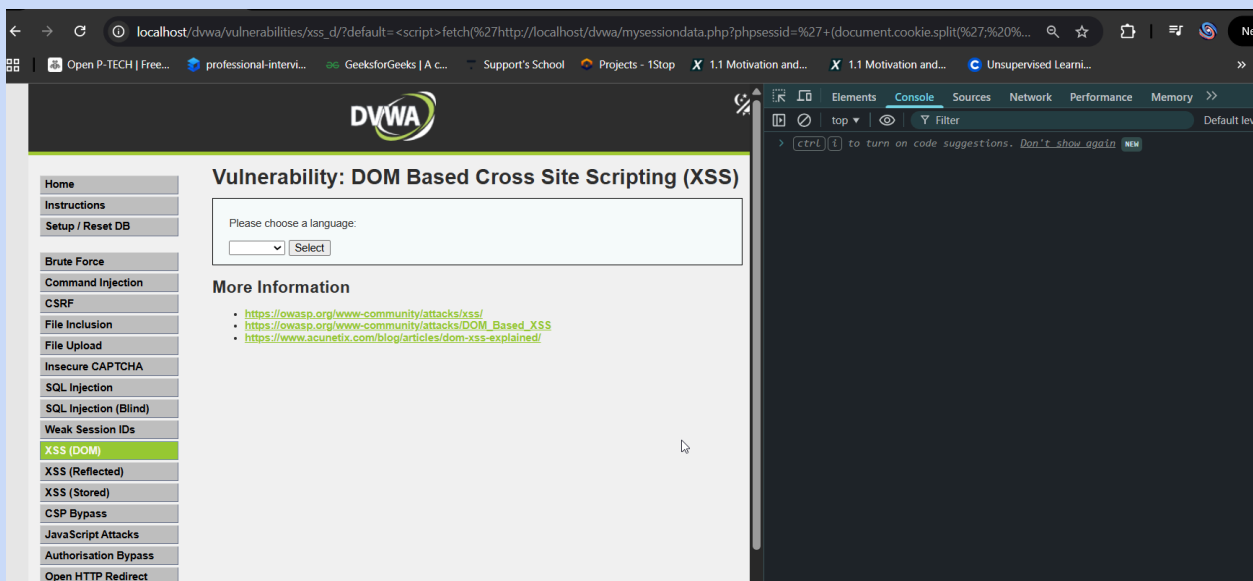
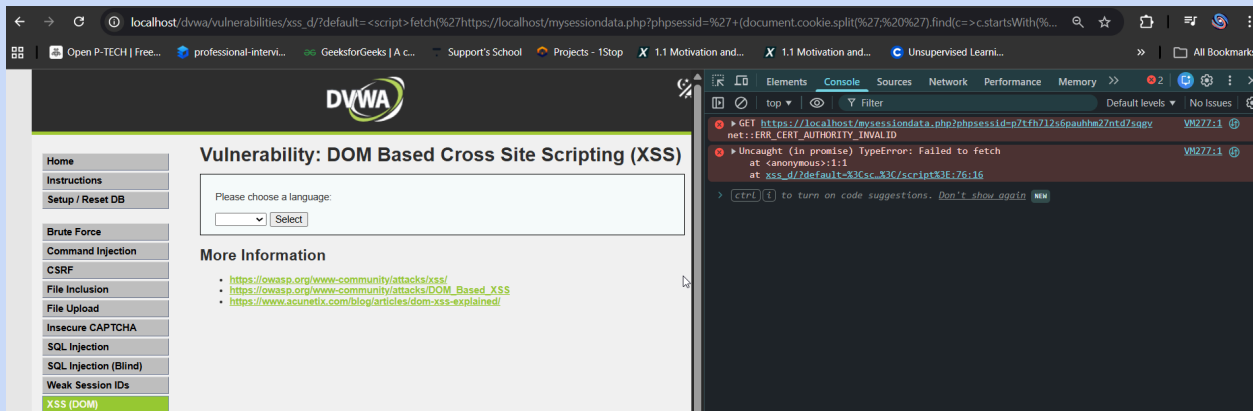
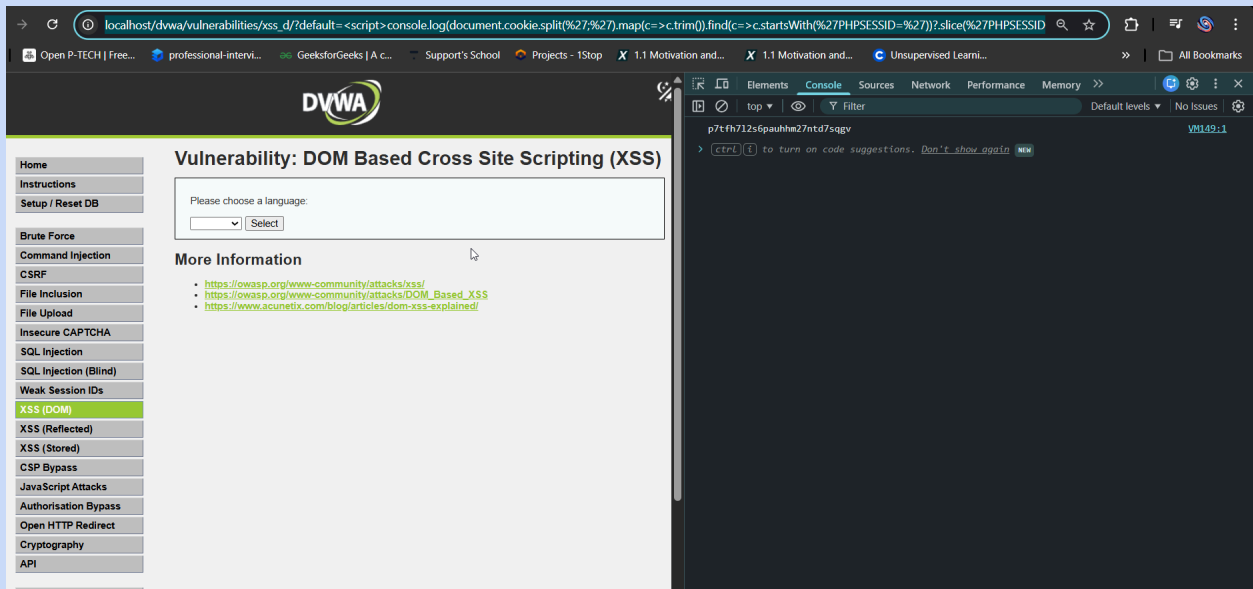
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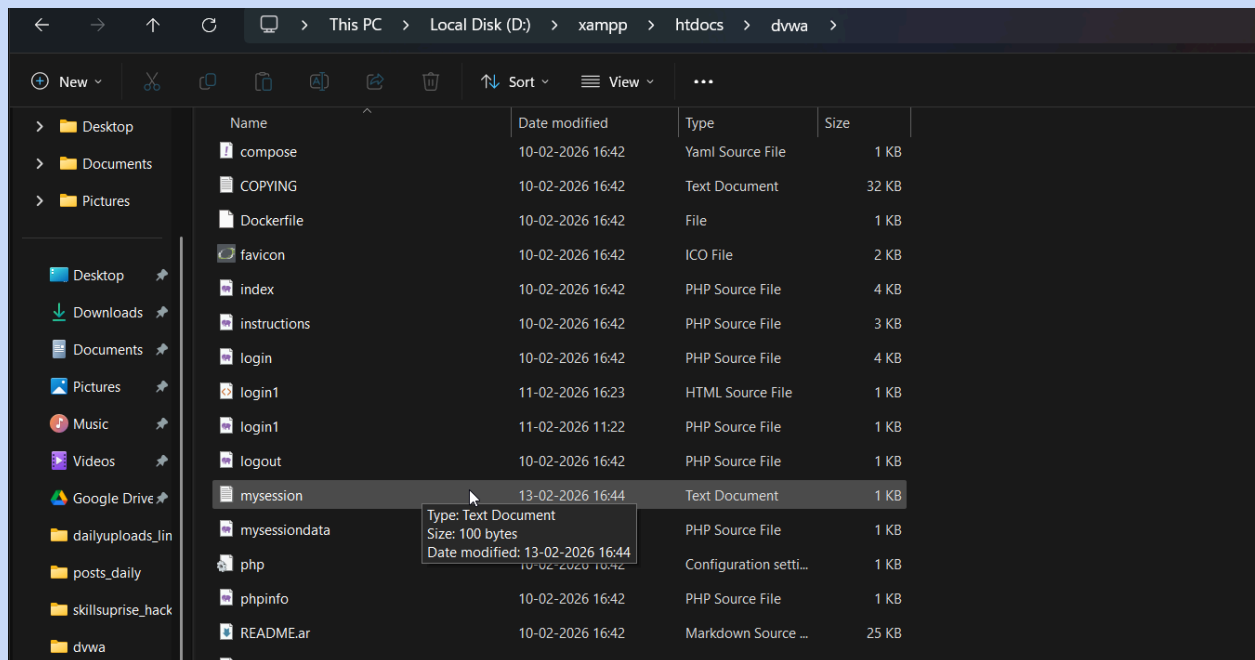
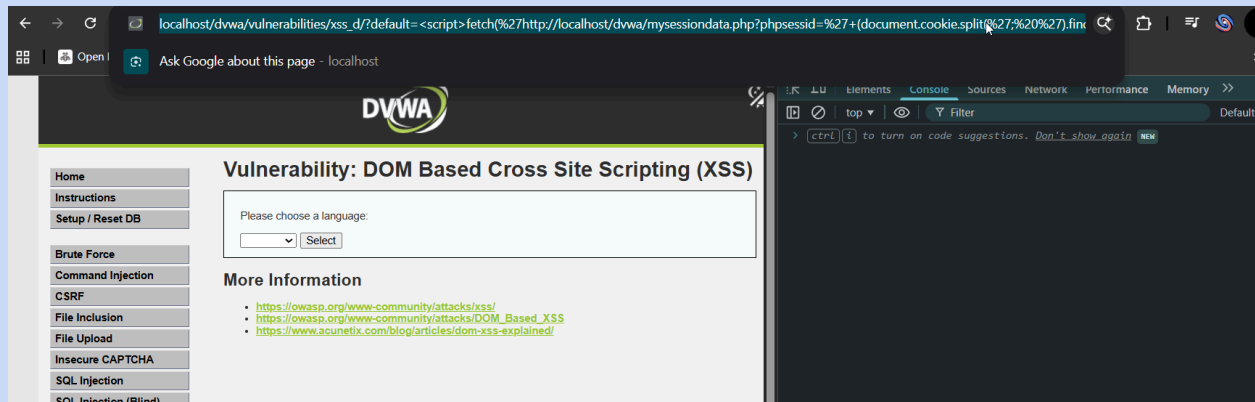
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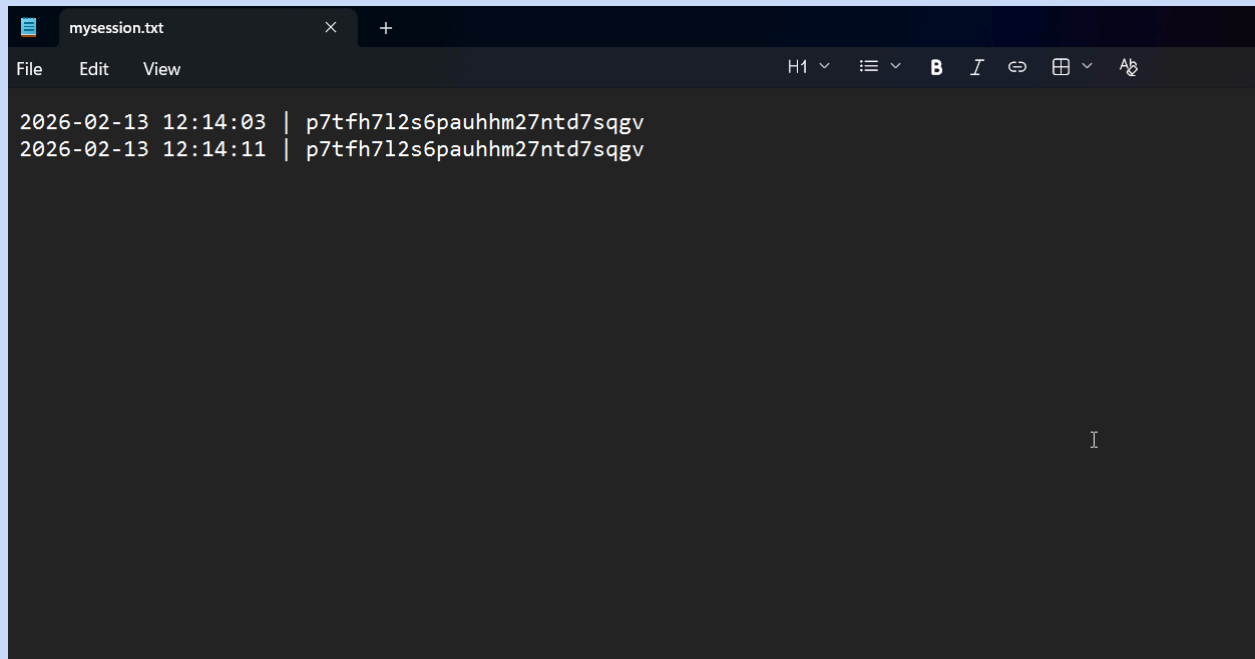
More Information

- <https://owasp.org/www-community/attacks/xss/>
- https://owasp.org/www-community/attacks/DOM_Based_XSS
- <https://www.acunetix.com/blog/articles/dom-xss-explained/>









A screenshot of a text editor window titled "mysession.txt". The editor has a dark theme and a menu bar with "File", "Edit", and "View". The toolbar includes options for heading (H1), list, bold (B), italic (I), link, table, and undo. The text content shows two log entries separated by a vertical bar:

```
2026-02-13 12:14:03 | p7tfh712s6pauhhm27ntd7sqgv  
2026-02-13 12:14:11 | p7tfh712s6pauhhm27ntd7sqgv
```

Summary:

JavaScript DOM vulnerabilities, especially DOM-based XSS, are client-side threats arising from unsafe data flows in JS, enabling attackers to execute code in browsers. Originating in 2005 amid rising client-side complexity, they've affected giants like Google and remain common (thousands reported yearly). Ethically, they're key for pentesting with tools like Burp and ZAP. They differ from server-side XSS by being browser-only, with severe impacts like data theft. Trends show AI exacerbating issues, but better frameworks and tools offer hope.